An Uncomplicated Overview

How Sapiens' New Dynamic Memory Revolutionizes Artificial Intelligence

Intelligent Organization, Instant Search and Zero-Cost Response Generation (SAPITensors)

Classification: Commercial and Product Documentation (General Public)

Abstract

This document translates the magic behind our innovative Artificial Intelligence architecture, *Dynamic Tensor Indexing*, created by Sapiens Technology®. Today, traditional AIs spend an absurd amount of energy "reading" an entire library every time you ask a question (a heavy and expensive process that requires gigantic video cards). Our system changes the rules of the game: we transformed the AI into a gigantic and perfect **organized file cabinet**. Instead of searching in the dark, the system knows exactly which "drawer" to open. This allows our AI to learn new things without forgetting the old ones and to answer your questions at the speed of light, spending almost zero battery. This inaugurates a new era where super-powerful AI can run even on simple computers, cell phones or devices without internet access.

1 The Art of Focusing on What Matters (Intelligent Attention)

Most AIs try to pay attention to all the words and noises in a text at once, which makes them slow and confused. Our technology rejects this heavy approach. Instead, we created an intelligent filter. Imagine a speed reader who can ignore "filler" words (like "a", "of", "to") and capture only the absolute essence of what you said.

1.1 Extracting the Gold from Information

Unlike common AIs (which get exponentially slower the longer the text they need to read), our algorithm uses pure logic to select only the main ideas. It's like using a powerful magnet to pull only the needles out of the haystack:

$$\text{Perfect Search} = \text{Core Ideas} - \text{Unnecessary Noise} \tag{1}$$

Once we find the core ideas, the order in which you say the words is no longer a problem. If you ask "What color is the sky?" or "The sky is what color?", our system understands that the **intention is the same**. This drastically reduces the computer's effort to understand what you want.

2 The Super File Cabinet: Organizing the AI's Mind

The true brilliance of our system lies in how it arranges its own "brain". Normal AIs store information in a large, messy warehouse. When you ask something, they search the entire warehouse. We solved this by creating what we call *Organized Spaces*.

To avoid wasted time searching through everything, knowledge is fractured into perfect categories. If your question is about sports, the AI will never look in the culinary drawer:

$$\text{Subject of your Question} \iff \text{Specific File Cabinet Drawer} \tag{2}$$

Mathematically, we divide the "warehouse" into thousands of small, independent boxes. When the system needs an answer, it goes straight to the right box. The result? A speed acceleration so great it seems like science fiction.

3 The End of Forgetting: An AI that Learns Forever

One of the biggest problems with modern AI is "catastrophic forgetting". If you teach a normal AI how to speak Spanish, and then try to teach it French, its internal "wiring" gets tangled and it starts to forget Spanish. The outdated concept of "training" the AI is flawed.

We replaced this with the construction of an expandable map. It's like building a house with Lego blocks: adding a new room doesn't knock down the old walls.

Proven Advantage 3.1 (Eternal and Accumulative Learning). *Let C be the total knowledge of the AI today. Our mathematical update rule ensures that anything new the AI learns tomorrow will only add to what it already knows:*

$$\textit{Tomorrow's Knowledge} \geq \textit{Today's Knowledge} \tag{3}$$

This guarantees that our model never suffers from interference. The learning is perpetual, safe and never erases the past.

4 The "Closest" Rule: Dealing with the Unknown

What happens when you ask the AI something it has never seen exactly in that way? Normal models tend to "hallucinate" (invent lies) or freeze. Our model doesn't panic. It uses the rule of familiarity.

If you ask for a "turquoise blue" shirt and the system doesn't have that exact word, it analyzes the options and gives you the closest color it has. It calculates the "distance" between your question and what it knows, taking the safest shortcut:

$$\text{Response} = \text{Search for the stored idea most similar to your question} \tag{4}$$

This mathematics of "approximation" makes the AI incredibly flexible. Difficult queries or unusual words are gently drawn to the closest correct concept that the AI already masters, avoiding nonsensical answers.

5 Maximum Speed and Zero-Cost Generation

The final impact of our architecture completely destroys the limitations of large AIs (like ChatGPT or similar). Currently, when an AI generates a long text, it needs to "think" from scratch for each new word it writes. That's why long texts take longer and cost fortunes in energy.

The inference of the Sapiens system works in a totally different way.

Proven Advantage 5.1 (Zero-Cost Text Generation). *Unlike common AIs that get increasingly tired and slow with each generated word, our system understands the whole idea in millisecond zero. After that, the effort to simply "print" the words on your screen drops to a value so low that we consider it constant:*

$$\textit{AI effort to generate words} = \textit{Practically Zero} \tag{5}$$

This radically decouples the size of your response from your device's battery drain.

This consecrates Sapiens Technology®'s architecture as the definitive and indisputable solution for running Artificial Intelligence in the real world: fast, never forgets, understands your intentions and works perfectly even on the simplest and cheapest devices on the market.